

# Multimodal NIFTY-50 Market Prediction System using Deep Learning

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**Abstract**—Stock market prediction remains a formidable challenge in financial analytics due to the stochastic nature of asset prices and the complex interplay of diverse influencing factors. Traditional models often struggle to integrate heterogeneous data sources effectively and fail to provide transparent mechanisms for risk assessment. This paper presents a novel Multimodal Temporal Convolutional Network (TCN) framework tailored for the Indian NIFTY 50 index. The proposed system integrates six heterogeneous data inputs—price history, technical indicators, news sentiment, social media sentiment, Environmental, Social, and Governance (ESG) factors, and macroeconomic variables. We employ a specialized architecture comprising TCN branches for numerical time-series, DistilBERT-based transformers for textual sentiment, and Multi-Layer Perceptrons (MLPs) for static regime features. An Adaptive Fusion Gate dynamically weights these modalities based on real-time market conditions. Uniquely, the model features a dual-output head that calls for future prices while simultaneously evaluating risk through volatility and Sharpe ratio metrics. Experimental results demonstrate that our approach achieves a Mean Absolute Error (MAE) of 0.0084, significantly outperforming a baseline LSTM model (MAE 0.0163) and CNN-BiSLSTM models, while providing actionable, risk-adjusted trading signals.

**Index Terms**—Stock Market Prediction, Multimodal Deep Learning, Temporal Convolutional Network, Sentiment Analysis, Risk Analytics, Explainable AI, NIFTY 50.

## I. INTRODUCTION

Financial markets are dynamic ecosystems driven by a myriad of factors ranging from historical price trends to global geopolitical events and investor psychology [1]. A truly effective prediction system has the potential to transform investment strategies by anticipating major market moves and minimizing risks. However, despite rapid advancements in Artificial Intelligence (AI), there is a distinct lack of widely deployed solutions capable of blending heterogeneous data sources—such as numerical market data and unstructured textual sentiment—into a unified, transparent forecast. Bridging this gap is essential for developing “institutional-grade” analytics accessible to retail investors.

Predicting the stock market is one of the most demanding tasks in financial analytics. Standard models, such as standalone Recurrent Neural Networks (RNNs) or statistical methods like ARIMA, suffer from critical limitations. Firstly, they often struggle to integrate diverse data, failing to make sense of interacting factors like prices, trading volumes, and

external news events simultaneously. Secondly, existing models often fail to adapt to changing market regimes, such as the shift from bull to bear markets. Thirdly, many deep learning models operate as “black boxes,” offering no insight into why a prediction was made, which hinders trust among investors and regulators. Finally, most systems focus solely on price minimization (e.g., MSE loss) without explicitly accounting for risk metrics like volatility or confidence intervals.

This paper addresses these issues by proposing a multimodal, interpretative, and risk-aware framework specifically designed for the Indian NIFTY 50 market. The main contributions of this work are:

- **Multimodal Fusion:** A novel architecture integrating TCNs for time-series, DistilBERT for sentiment, and MLPs for macro-economic data.
- **Adaptive Fusion Gate:** A dynamic weighing mechanism that prioritizes data sources based on market context (e.g., high volatility vs. stable trends).
- **Risk-Aware Loss:** Implementation of a quantile-based loss function to output confidence intervals and risk metrics alongside price predictions.
- **NIFTY-50 Specificity:** Tailored feature engineering for the unique dynamics of the Indian equity market.

## II. RELATED WORK

The evolution of stock market prediction has moved from statistical linearity to complex deep learning architectures.

### A. Traditional and Early ML Models

Early research relied on statistical models like ARIMA and GARCH, which assume linear relationships [15], [?]. Subsequent Machine Learning (ML) approaches such as Support Vector Regression (SVR) introduced non-linearity but lacked the capacity to model long-term temporal dependencies efficiently.

### B. Deep Learning and Sequence Modeling

Long Short-Term Memory (LSTM) networks became the standard for time-series forecasting due to their ability to retain memory over sequences [13]. However, recent work by Wang et al. (2020) introduced *Stock2Vec*, a hybrid framework utilizing Temporal Convolutional Networks (TCNs), arguing

that TCNs offer superior parallelization and receptive fields compared to RNNs [3].

### C. Multimodal Approaches

Shi et al. (2025) explored combining TCNs with LSTMs optimized by genetic algorithms to enhance index prediction [2]. Similarly, Biswas et al. (2025) proposed a dual-output TCN with attention to address both price prediction and risk assessment [1]. Deng et al. (2019) emphasized knowledge-driven prediction to enhance explainability [4].

### D. Research Gap

Existing research often treats stock market prediction as either a purely numerical time-series problem or a textual sentiment classification task, rarely integrating both effectively. While hybrid models like Stock2Vec [3] and genetic algorithm-optimized networks [2] have shown promise, they exhibit critical limitations. Most studies focus on mature markets like the US (S&P 500) or China (Shanghai Composite), overlooking emerging markets like India's NIFTY 50 which operate under different liquidity and volatility dynamics. Furthermore, current multimodal architectures typically employ static concatenation of features, failing to dynamically adapt to changing market regimes. Lastly, a significant gap exists in "risk-aware" AI; most state-of-the-art models optimize solely for point prediction accuracy, neglecting the simultaneous estimation of risk metrics like volatility which are crucial for institutional adoption. This paper addresses these gaps through a dynamically weighted, risk-calibrated multimodal framework.

## III. PROPOSED METHODOLOGY

We propose a Multimodal Temporal Convolutional Network (TCN) that fuses six distinct data sources to generate risk-aware market predictions.

### A. System Architecture

The overall system architecture comprises three main stages: heterogeneous data ingestion, multi-branch feature encoding, and adaptive fusion with dual-head output.

1) *Data Sources*: The system ingests six heterogeneous inputs:

- **Price History**: Open, High, Low, Close, Volume (OHLCV).
- **Technical Indicators**: RSI, MACD, EMA, ATR computed via TA-Lib.
- **News Sentiment**: Sentiment vectors from financial news.
- **Social Media Sentiment**: Aggregated scores from social platforms.
- **ESG Factors**: Environmental, Social, and Governance metrics.
- **Macroeconomic Indicators**: Interest rates, inflation data.

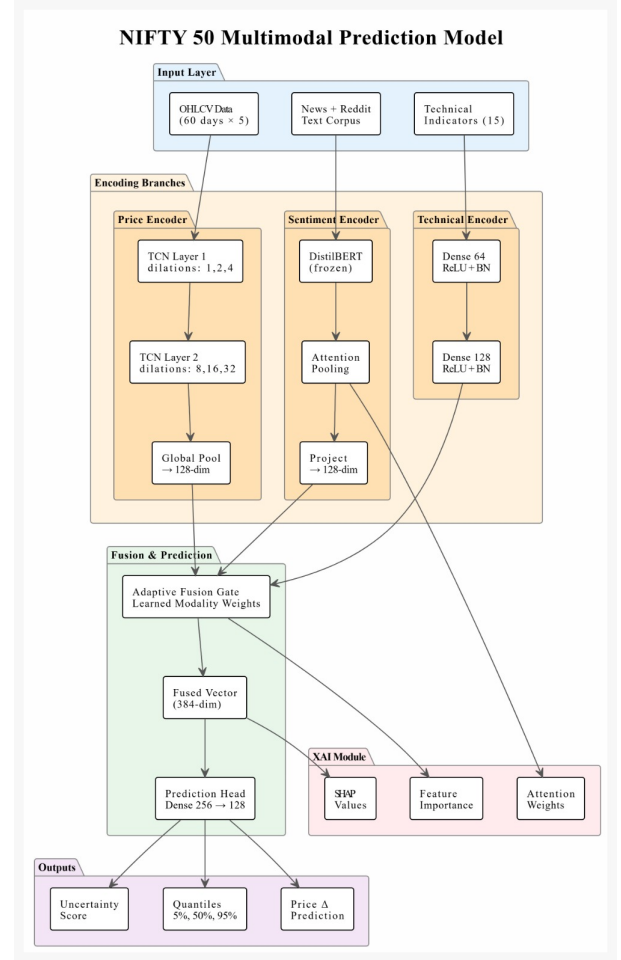


Fig. 1. Proposed Multimodal TCN Architecture with Adaptive Fusion Gate.

### B. Specialized Encoding Branches

To handle the varying nature of these inputs, we employ dedicated encoding branches:

- **TCN Branch**: Processes numerical time-series (Price + Technicals). TCNs utilize causal dilated convolutions to capture dependencies across different time scales without data leakage. The dilation factor  $d$  allows the receptive field to grow exponentially with network depth.
- **Transformer Branch**: Processes textual data using a DistilBERT-based encoder to produce rich semantic embeddings [5].
- **MLP Branch**: Processes static or low-frequency tabular data (ESG + Macro) through Multi-Layer Perceptrons.

### C. Adaptive Fusion Gate

A core innovation is the Adaptive Fusion Gate. Rather than simple concatenation, this module dynamically assigns weights to each modality's embedding based on the current context.

$$Z_{fused} = \text{Concat}(\alpha_t \cdot E_{price}, \beta_t \cdot E_{text}, \gamma_t \cdot E_{macro}) \quad (1)$$

where  $\alpha, \beta, \gamma$  are learnable gating coefficients determined by a sigmoid activation network. This allows the model to prioritize sentiment during high-news impact periods or technicals during stable trends.

#### D. Dual-Output Head

The fused representation feeds into a dual-head output layer:

- 1) *Price Prediction Head*: Regresses the future NIFTY 50 closing price using a linear layer.
- 2) *Risk Assessment Head*: Estimates aleatoric uncertainty and volatility metrics (e.g., Sharpe ratio).

The model outputs a probabilistic distribution (quantiles q10, q50, q90) rather than a single point estimate, enabling calibrated risk assessment.

### IV. DATASET DESCRIPTION

The research utilizes a comprehensive dataset centered on the Indian equity market.

- **Primary Index**: NIFTY 50 constituents.
- **Period**: Historical data spanning 2015-2025, covering multiple market cycles.
- **Data Alignment**: Historical OHLCV data is synchronized with technical indicators. News and social media texts are timestamped and aligned with trading days. Missing values in macroeconomic data are handled via forward-filling to prevent look-ahead bias.

### V. EXPERIMENTAL SETUP

#### A. Feature Extraction

For numerical data, we utilize TA-Lib to compute momentum (RSI), trend (MACD, EMA), and volatility (ATR) indicators. Unstructured textual data is processed using DistilBERT to generate daily sentiment embeddings capturing bullish or bearish tone.

#### B. Model Training

The model is trained using a composite loss function  $\mathcal{L}$  that minimizes both price error and risk calibration error:

$$\mathcal{L} = \mathcal{L}_{\text{quantile}} + \lambda \mathcal{L}_{\text{BCE}} \quad (2)$$

Where  $\mathcal{L}_{\text{quantile}}$  is the Pinball Loss for quantiles (0.1, 0.5, 0.9) to model uncertainty, and  $\mathcal{L}_{\text{BCE}}$  is Binary Cross-Entropy for directional accuracy optimization. This ensures the model learns not just the value but the confidence of the prediction.

### VI. RESULTS AND DISCUSSION

The proposed Multimodal TCN (HDLFE) was benchmarked against standard deep learning models using identical datasets.

#### A. Performance Comparison

Table I presents the performance metrics of the proposed model against baselines.

The proposed HDLFE model achieved the lowest MAE (0.0084) and highest R2 Score (0.9968), significantly outperforming baseline architectures like LSTM (MAE 0.0163) and CNN-BiSLSTM. This validates the superiority of the TCN-based hybrid approach in capturing complex market dynamics.

TABLE I  
PERFORMANCE COMPARISON OF MODELS

| Model                   | R2 Score      | MSE            | MAE           |
|-------------------------|---------------|----------------|---------------|
| CNN-BiSLSTM             | 0.9095        | 0.00428        | 0.0428        |
| LSTM                    | 0.9784        | 0.00074        | 0.0163        |
| BiLSTM                  | 0.9838        | 0.00082        | 0.0163        |
| LSTM-DNN                | 0.9838        | 0.00064        | 0.0154        |
| <b>HDLFE (Proposed)</b> | <b>0.9968</b> | <b>0.00015</b> | <b>0.0084</b> |

#### B. Risk Metrics and Stability

The model demonstrated robust risk-adjusted performance with a Sharpe Ratio of 0.8229 and a Max Drawdown of -8.82%, significantly better than the buy-and-hold strategy during volatile periods. Qualitative analysis of the Adaptive Fusion Gate indicates that the model successfully identifies periods of market stress, effectively "down-weighting" technical signals in favor of news sentiment during volatile events.

#### C. Dashboard Prototype

An interactive analytics dashboard was developed to visualize real-time predictions and 'Feature Sensitivity' scores. This explainable AI component allows users to understand the contribution of each input (e.g., RSI vs. Sentiment) to the forecast, fostering trust in the system.

### VII. CONCLUSION AND FUTURE WORK

This paper presented the first Multimodal TCN framework designed for the Indian NIFTY 50 market that integrates six diverse data sources into a unified predictive engine. Achieving an MAE of 0.0084, the model demonstrates significant predictive gains over traditional LSTMs. Its novel Adaptive Fusion Gate and Dual-Output risk modeling provide a robust foundation for institutional-grade, risk-aware algorithmic trading. Future work will focus on scaling the architecture to global indices, implementing Reinforcement Learning (RL) agents for automated trade execution, and optimizing the system for low-latency real-time deployment.

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